

that accepts all the render buffers (noisy radiance, depth, normals, albedo, etc.) is trainable. Our encoder adds information to feature maps at varying spatial resolutions to the decoder block of the base diffusion model. **Dataset** We train and evaluate on our own procedurally-generated dataset of approx. 45k 256-res frames rendered from PBRT [10], as well as the open-source noisebase dataset [5]. These datasets consist of random arrangements of objects, camera pose, texture, and lighting, rendered with a path tracer at varying sampling rates and auxiliary buffers. We compare with existing SOTA works. We retrain Isik [7] and AFGSA [13] on our dataset, and use the pretrained benchmark OIDN [1]. Similar to other works, we apply log tonemapping to ensure high dynamic range is preserved.

Results

We present a qualitative evaluation in Fig 1. Notice that our approach synthesizes clean lines in the undersampled regions, whereas existing works, which do not have a notion of what a real image looks like, struggle to reconstruct these details. We then present a quantitative evaluation in Table 1. Across a range of standard benchmarks, our method outperforms existing work. Our method requires approx. 2.8 seconds to denoise a 256x256 image and 63 seconds to denoise an HD 1080x1920 frame (without skipping time steps) on an A40 GPU. We use mixed-precision during inference and the SUPER27 DDPM sampler in DeepFloyd. Even though diffusion is more expensive than single-pass methods, the cost of denoising remains much lower than the cost of rendering real-world scenes to convergence, which can be dozens of hours. Even scenes that fit into GPU memory can take dozens of minutes to render to convergence, which dwarfs the cost of denoising low-ray estimates. Future work may consider temporal coherence and few-step diffusion models.

Table 1: **Quantitatively, our method is competitive with SOTA.** We compare error metrics across different spp settings (4, 16, 64) for various methods on our test set, consisting of 226 sequences of 8 frames each at 256 res. All methods are run in single-frame mode and evaluation metrics do not take into account temporal performance. Our method outcompetes all others in L1, which is computed in HDR [0 – 6] space, as well as FoVVD. We are also competitive in other metrics, which evaluate both pixel-wise and perceptual quality. DINO [6, 9] and CLIP measure cosine similarity of the generated and GT global image feature vector. In the **fourth row**, we disable our Control Module and denoise with an off-the-shelf DeepFloyd Stage II model, presenting the best numbers after ablating the number of skipped denoising time steps and the *aug_level* parameter. Applying pure DeepFloyd to low-ray estimates is not successful. **Fifth row** Thus, conditioning our method with render buffer information via our Control Module makes a significant difference to performance.

Method	L1 ↓	PSNR ↑	LPIPS ↓	DINO ↑ [9]	CLIP ↑ [11]	FliP ↓ [3]	FoVVD ↑ [8]
AFGSA [13]	0.0279	38.672	0.1130	0.914	0.932	0.0494	8.699
Isik [7]	0.0499	38.828	0.0871	0.945	0.955	0.0476	8.835
OIDN [1]	0.0638	36.329	0.1192	0.915	0.904	0.0691	8.645
DeepFloyd-II [2]	0.1009	27.583	0.3860	0.742	0.804	0.1390	6.513
Ours	0.0237	39.130	0.0748	0.948	0.965	0.0487	8.888

4spp

Method	L1 ↓	PSNR ↑	LPIPS ↓	DINO ↑ [9]	CLIP ↑ [11]	FliP ↓ [3]	FoVVD ↑ [8]
AFGSA [13]	0.0184	42.076	0.0708	0.939	0.958	0.0350	9.222
Isik [7]	0.0449	42.044	0.0560	0.964	0.972	0.0346	9.283
OIDN [1]	0.0537	39.869	0.0858	0.937	0.924	0.0459	9.158
DeepFloyd-II [2]	0.0825	28.988	0.350	0.775	0.819	0.1076	7.150
Ours	0.0156	42.343	0.0499	0.965	0.975	0.0338	9.328

16spp

Method	L1 ↓	PSNR ↑	LPIPS ↓	DINO ↑ [9]	CLIP ↑ [11]	FliP ↓ [3]	FoVVD ↑ [8]
AFGSA [13]	0.0142	45.091	0.0467	0.956	0.972	0.0260	9.562
Isik [7]	0.0433	45.055	0.0341	0.977	0.982	0.0260	9.593
OIDN [1]	0.0488	42.664	0.0640	0.954	0.940	0.0329	9.482
DeepFloyd-II [2]	0.0696	29.777	0.317	0.786	0.830	0.0973	7.261
Ours	0.0113	44.953	0.0346	0.975	0.982	0.0261	9.616

64spp

References

- [1] Attila T. Áfra. Intel® Open Image Denoise, 2024. <https://www.openimagedenoise.org>
- [2] Stability AI. Deepfloyd if. <https://github.com/deep-floyd/IF>
- [3] Pontus Andersson, Jim Nilsson, Tomas Akenine-Möller, Magnus Oskarsson, Kalle Åström, and Mark D. Fairchild. FLIP: A Difference Evaluator for Alternating Images. *Proceedings of the ACM on Computer Graphics and Interactive Techniques*, 3(2):15:1–15:23, 2020.
- [4] Steve Bako, Thijs Vogels, Brian McWilliams, Mark Meyer, Jan Novák, Alex Harvill, Pradeep Sen, Tony DeRose, and Fabrice Rousselle. Kernel-predicting convolutional networks for denoising monte carlo renderings. *ACM Transactions on Graphics (TOG) (Proceedings of SIGGRAPH 2017)*, 36(4), 2017.
- [5] Martin Balint, Krzysztof Wolski, Karol Myszkowski, Hans-Peter Seidel, and Rafał Mantiuk. Neural partitioning pyramids for denoising monte carlo renderings. In *ACM SIGGRAPH 2023 Conference Proceedings*, New York, NY, USA, 2023. Association for Computing Machinery.
- [6] Timothée Darcet, Maxime Oquab, Julien Mairal, and Piotr Bojanowski. Vision transformers need registers, 2023.
- [7] Mustafa Işık, Krishna Mullia, Matthew Fisher, Jonathan Eisenmann, and Michaël Gharbi. Interactive monte carlo denoising using affinity of neural features. *ACM Trans. Graph.*, 40(4), 2021.
- [8] Rafał K. Mantiuk, Gyorgy Denes, Alexandre Chapiro, Anton Kaplanyan, Gizem Rufo, Romain Bachy, Trisha Lian, and Anjul Patney. Fovvideovdp: a visible difference predictor for wide field-of-view video. *ACM Trans. Graph.*, 40(4), 2021.
- [9] Maxime Oquab, Timothée Darcet, Theo Moutakanni, Huy V. Vo, Marc Szafraniec, Vasil Khalidov, Pierre Fernandez, Daniel Haziza, Francisco Massa, Alaaeldin El-Nouby, Russell Howes, Po-Yao Huang, Hu Xu, Vasu Sharma, Shang-Wen Li, Wojciech Galuba, Mike Rabbat, Mido Assran, Nicolas Ballas, Gabriel Synnaeve, Ishan Misra, Herve Jegou, Julien Mairal, Patrick Labatut, Armand Joulin, and Piotr Bojanowski. Dinov2: Learning robust visual features without supervision, 2023.
- [10] Matt Pharr, Wenzel Jakob, and Greg Humphreys. *Physically Based Rendering: From Theory to Implementation*. Morgan Kaufmann, 2016.
- [11] Alec Radford, Jong Wook Kim, Chris Hallacy, Aditya Ramesh, Gabriel Goh, Sandhini Agarwal, Girish Sastry, Amanda Askell, Pamela Mishkin, Jack Clark, Gretchen Krueger, and Ilya Sutskever. Learning transferable visual models from natural language supervision, 2021.
- [12] Chitwan Saharia, William Chan, Saurabh Saxena, Lala Li, Jay Whang, Emily Denton, Seyed Kamyar Seyed Ghasemipour, Burcu Karagol Ayan, S. Sara Mahdavi, Rapha Gontijo Lopes, Tim Salimans, Jonathan Ho, David J Fleet, and Mohammad Norouzi. Photorealistic text-to-image diffusion models with deep language understanding, 2022.
- [13] Jiaqi Yu, Yongwei Nie, Chengjiang Long, Wenju Xu, Qing Zhang, and Guiqing Li. Monte carlo denoising via auxiliary feature guided self-attention. *ACM Trans. Graph.*, 40(6), 2021.
- [14] Lvmin Zhang, Anyi Rao, and Maneesh Agrawala. Adding conditional control to text-to-image diffusion models, 2023.